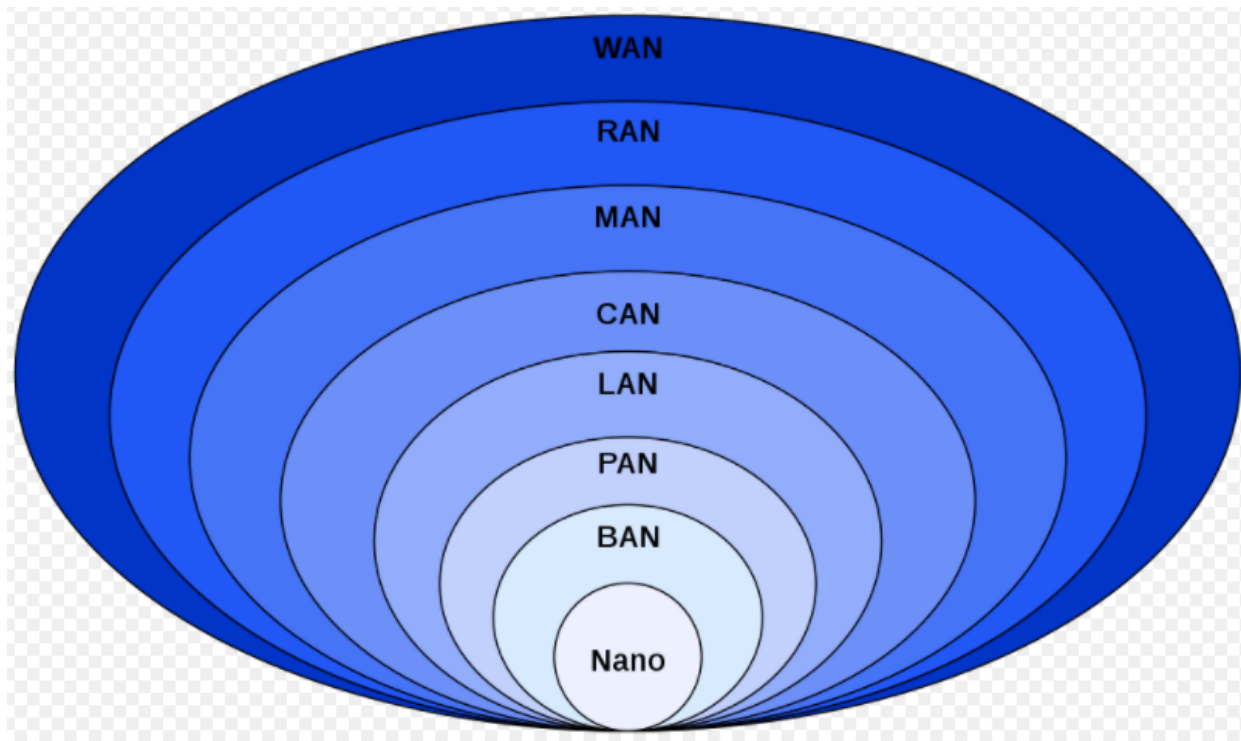


Computer Network Types

A computer network is a group of computers linked to each other that enables the computer to communicate with another computer and share their resources, data, and applications.



- LAN(Local Area Network)
- PAN(Personal Area Network)
- MAN(Metropolitan Area Network)

- **WAN(Wide Area Network)**
- **SAN(Storage Area Network)**
- **NAN(Near-me area network)**
- **BAN(Body Area Network)**
- **CAN(campus area network)**
- **RAN(Radio area network)**

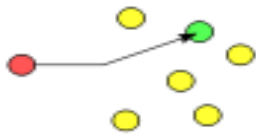
Routing schemes differ in how they deliver messages:

- **Unicast** delivers a message to a single specific node using a *one-to-one* association between a sender and destination: each destination address uniquely identifies a single receiver endpoint.
- **Broadcast** delivers a message to all nodes in the network using a *one-to-all* association; a single datagram from one sender is routed to all of the possibly multiple endpoints associated with the broadcast address. The network automatically replicates datagrams as needed to reach all the recipients within the scope of the broadcast, which is generally an entire network subnet.

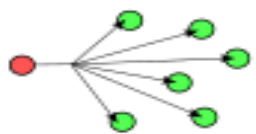
- **Multicast** delivers a message to a group of nodes that have expressed interest in receiving the message using a *one-to-many-of-many* or *many-to-many-of-many* association; datagrams are routed simultaneously in a single transmission to many recipients. Multicast differs from broadcast in that the destination address designates a subset, not necessarily all, of the accessible nodes.
- **Anycast** delivers a message to any one out of a group of nodes, typically the one nearest to the source using a *one-to-one-of-many* association where datagrams are routed to any single member of a group of potential receivers that are all identified by the same destination address. The routing algorithm selects the single receiver from the group based on which is the nearest according to some distance measure.
- **Geocast** delivers a message to a group of nodes in a network based on their geographic location. It is a specialized form of multicast addressing used by some routing protocols for mobile ad hoc networks.

Routing schemes

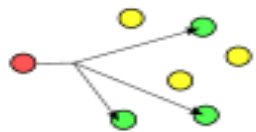
Unicast



Broadcast



Multicast



Anycast



Geocast



